

MATTHEW D. HALL

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EDUCATION

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PhD - In Progress - *The University of Sheffield* OCT 2017 - PRESENT

PROJECT: **Self-Reconfiguration of Modular Robots via Active Subtraction**
Developing and verifying novel control algorithms for the autonomous reconfiguration of multi-robot modular systems, enabling modules to actively remove themselves from a starting configuration to form a desired shape.

SUPERVISORS: Dr Roderich Gross, Dr Shuhei Miyashita.

AFFILIATIONS: Natural Robotics Lab, Sheffield Robotics, Department of Automatic Control & Systems Engineering.

SCHOLARSHIPS: TUOS and EPSRC Doctoral Training Partnership National Productivity Scholarship for course and maintenance fees.
University Postgraduate Research Student Publication Scholarship for 3 months funding towards journal publication.

Master of Engineering (MEng) - *Lancaster University* OCT 2012 - JUL 2016

SUBJECT: **Electrical and Electronics Engineering - 1st Class Hons**

PROJECTS: Spatial Distribution of Blood Flow in Peripheral Vein Networks by Non-Invasive Near-Infrared Imaging. Design and Construction of an Optical Pickup System for an Electric Bass Guitar.

SUPERVISORS: Dr Farid Aiouache, Professor Graeme Burt.

EXPERIENCE

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Peer Reviewer - *Natural Robotics Lab* OCT 2017 - PRESENT

Review submitted works for a number of highly regarded international conferences, such as: AAMAS, ANTS, DARS, IROS, MRS, RSS and TAROS. Aids with keeping up to date with state of the art work. Utilises my ability to disseminate information and assess the quality of written works.

Graduate Teaching Assistant - *University of Sheffield* OCT 2017 - OCT 2020

Assisted in the planning and running of laboratory based teaching and seminar sessions for Master's and Bachelor's students. Required an in-depth knowledge of the material and the ability to clearly communicate to answer the students' questions. Organisational skills also required to effectively plan sessions for multiple groups for the semester.

Tourism Advisor - *Use-it Brussels* JAN 2017 - AUG 2017

Part-time work within a sustainable tourism project while residing in Belgium, offering tourist information and running tours of the lesser-known parts of Brussels. A great opportunity to thoroughly learn about the city and its culture, as well as improving my interpersonal skills.

Korfball Team Captain & Coach - Lancaster Ligers OCT 2015 - JUN 2016

Earned Level 2 coaching and referee qualification in Aug 2015. Assisted with training and squad selection. Also responsible for fixture and travel arrangements. Invaluable in furthering team-leadership skills.

Student Ambassador - Lancaster University OCT 2014 - JUN 2016

Part-time employment assisting with university events and the regular running of tours around the university campus. These responsibilities required excellent communication and organisational skills, and a personable yet professional demeanour.

PUBLICATIONS

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M. D. Hall, A. Özdemir, R. Groß, **“Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots”**, in *Proc. Robotics: Science and Systems XVI (RSS)*, RSS Foundation, 2020.

A. Özdemir, M. Gauci, A. Kolling, **M. D. Hall**, R. Groß, **“Spatial Coverage Without Computation”**, in *Proc. 2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pages 9674–9680.

PRESENTATIONS

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2021: Departmental Postgraduate Research Symposium - *Virtual* - **Awarded “Best Oral Presentation”**.

2020: Int’l Conf. on Robotics: Science and Systems - *Virtual* - available at https://youtu.be/RgGi_mc4t0Q.

2019: Self-Configuring Modular Robotics for Earth and Space - *York, UK*.
IEEE Int’l. Conf. on Robotics and Automation - *Montreal, Canada*
UK & I Conf. on Transformative Robotics for Industries - *London, UK*.
Faculty Researcher Symposium - *Sheffield, UK* - **Awarded “Best Poster Presentation”**.

SKILLS & INTERESTS

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PROGRAMMING: Proficient in **Python**, **C/C++**, **Git**, **LaTeX**, **Shell Scripting**.

Working knowledge of **MATLAB**, **Java**, **HTML**, **CSS** and **VHDL**.

Designed and created personal website using **Hugo** framework.

Released a small number of customisation apps for **Android**.

Selection of other projects available on my **GitLab**.

MUSIC: Bassist and guitarist. Experience in Ableton music production.
Regular guest radio DJ for “Clear A Space” on Gaga Radio.

FILM: Keen interest in film. Written, shot and edited amateur films with family and friends.

OTHER: Food and cooking, games (board or video), craft beer.

REFERENCES AVAILABLE UPON REQUEST

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